

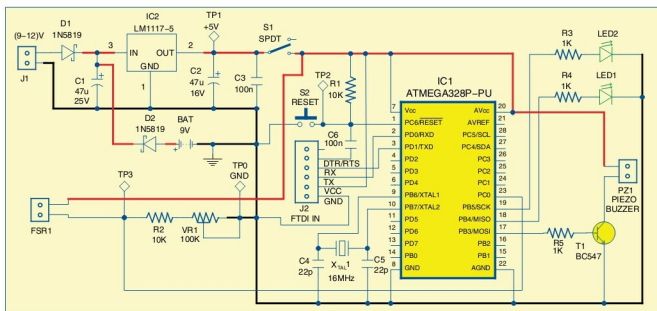


Fig. 2: Circuit diagram of the bed-vacancy alarm system

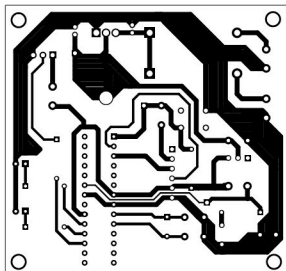


Fig. 3: Actual-size PCB pattern of the bed-vacancy alarm system

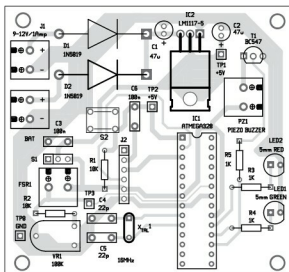


Fig. 4: Component layout of the PCB

larger beds, deploy several FSR sensors, so that the bed is assumed to be empty when none of the sensor values

are above the threshold limit.

Software program

The software has the following functions:

```
void setup(void)
```

This function is used to initialise control pins as digital output.

```
void loop(void)
```

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

This function is used to read FSR output and generate an alarm, depending on the threshold value.

In case a USB interface is absent in the system board, it becomes necessary to use a home-made/ready-made USB-to-serial converter for

programming the MCU. Inexpensive and easy-to-use USB-to-serial converters (FTDI basic boards) are now widely-available. Usually, one USB interface connection and six output connections (DTR, RXI, TXO, 5V, CTS and GND) are available in FTDI basic. Connect the circuit as shown in Table II (not tested in EFY Lab). ●



T.K. Hareendran is an electronics hobbyist, freelance technical writer and circuit designer